

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY,
BELGAUM, KARNATAKA**



**PROJECT REPORT
ON
DESIGN AND DEVELOPMENT OF FINANCIAL MARKETS
ANALYSIS TOOL**

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AWARD OF THE DEGREE OF**

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COMPUTER SCIENCE AND ENGINEERING**

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CERTIFICATE

This is to certify that the project titled **Design And Development Of Financial Markets Analysis Tool** is a bonafide work carried out by **Anuj Savant (2SD23CS014), Dhareesh Dodmani(2SD23CS030), Harshit Sommanvar (2SD23CS036), Mahant S Arawal (2SD23CS052)** submitted for the award of the degree of **BACHELOR OF ENGINEERING in COMPUTER SCIENCE AND ENGINEERING AND TECHNOLOGY, DHARWAD,KARNATAKA** (An autonomous institution affiliated to Visvesvaraya Technological University, Belgaum, Karnataka), during the year 2025-2026. It is certified all corrections /suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The project has been approved, as it satisfies the academic requirements in respect of project report prescribed for the said degree.

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Name of Examiners

Signature with Date

1)

2)

DECLARATION

We hereby declare that the dissertation work titled **Design And Development Of Financial Markets Analysis Tool**, has been carried out under the guidance of **Dr. Raghvendra G.S, Assistant Professor, Department of Computer Science and Engineering, S.D.M. College of Engineering and Technology, Dharwad**, in fulfillment of the degree of **Bachelor of Engineering in Computer Science and Engineering** from **Visvesvaraya Technological University, Belgaum, Karnataka**, during the academic year 2025-2026.

I also declare that I have not submitted this dissertation to any other university for the award of any other degree.

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ABSTRACT

The increasing need for accuracy, efficiency, and transparency in academic administration has led to the adoption of digital solutions for routine institutional activities. One such critical task is attendance monitoring, which traditionally relies on manual entry in registers or spreadsheets. These conventional methods are time-consuming, error-prone, and provide limited analytical insights. To address these limitations, this project focuses on the **Design And Development of Financial Markets Analysis Tool** that automates the entire attendance-tracking process while ensuring reliability, scalability, and ease of use.

The proposed system integrates a user-friendly interface with a secure backend to enable instructors to record, manage, and analyze attendance with minimal effort. Core features include student enrollment management, subject-wise attendance recording, real-time attendance percentage calculation, and automated report generation. The system architecture is built using modular design principles, allowing seamless integration of additional features such as notifications, analytics dashboards, and biometric or RFID inputs in the future. Validation rules are incorporated to reduce human error, while secure authentication ensures data privacy for both teachers and students.

This system significantly enhances productivity by reducing manual workload, improving accuracy, and enabling quick retrieval of attendance records for academic reviews, internal assessments, and performance monitoring. The project demonstrates the practical application of software engineering principles—including requirement analysis, system design, database modeling, and user interface development—to solve real-world academic management challenges. Overall, the Classroom Attendance Management System offers an efficient, reliable, and scalable digital solution capable of supporting institutions in modernizing their administrative workflows.

PROBLEM STATEMENT

Attendance monitoring is a routine but essential activity in educational institutions, as it directly influences academic performance evaluation, eligibility criteria, and overall student discipline. Despite its importance, many institutions still rely on traditional manual methods such as paper registers or spreadsheet entries to record attendance. These conventional practices present several challenges including high chances of human error, difficulty in maintaining large volumes of records, time-consuming processes, and lack of real-time accessibility. Manual tracking also makes it difficult to quickly generate reports, identify absenteeism patterns, or retrieve historical attendance data when required for audits, assessments, or administrative decisions.

Furthermore, as class sizes grow and academic requirements become more complex, manual methods fail to provide reliability, accuracy, or scalability. Issues such as misplaced registers, illegible handwriting, unauthorized alterations, and data redundancy further compromise the integrity of the attendance process. Teachers often spend a significant portion of class time marking attendance, reducing the time available for teaching and interaction. Additionally, administrators lack an efficient mechanism to monitor attendance trends across multiple classes, departments, or semesters.

To address these limitations, there is a clear need for a **Digital, Automated, Face Authenticated Classroom Attendance Management System** that streamlines the entire process. The system should enable fast, accurate, correct face matching and secure attendance recording; ensure consistent storage and retrieval of data; generate analytical reports in various formats like PDF, CSV; and support easy monitoring by Student and Teachers. Such a system must be user-friendly, reliable, and capable of handling institutional-scale data while minimizing manual workload and ensuring data integrity. The problem, therefore, lies in developing a robust, efficient, and scalable digital solution that replaces outdated manual attendance practices with a modern, integrated, face driven and technologically sound management system.

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1.1 Footnotes

- **Attendance accuracy is one of the primary indicators of academic performance**, and institutions globally emphasize its proper recording to ensure discipline and learning engagement.
- Traditional **paper-based attendance registers are prone to human error**, time delays, manipulation, and physical damage, making them inefficient for modern academic environments.
- **Automated attendance systems significantly reduce operational workload**, especially for teachers handling multiple classes and large student batches.
- **Digital attendance solutions support real-time data availability**, enabling faculty and administrators to monitor attendance without waiting for manual tabulation or end-of-semester calculations.
- **Software-based attendance systems also help maintain uniformity in institutional policies**, as standardized rules (e.g., minimum percentage criteria) can be implemented programmatically.
- **The development of such a system often follows the Software Development Life Cycle (SDLC)**—most commonly the Waterfall or Agile model—to ensure systematic planning, design, implementation, and testing.
- **Database management systems (DBMS)** form the backbone of this classroom attendance system, storing essential information such as **student profiles, face embeddings, classroom details, teacher data, daily attendance logs, and published notices**. MongoDB (a NoSQL DB) is used because it allows flexible document-based storage, which is ideal for handling face-recognition data, dynamic classroom structures, and rapidly changing attendance records.
- **Unlike relational databases, MongoDB does not require fixed tables or schemas**. This provides advantages for your project, such as storing:

- multiple face embeddings per student,
 - daily attendance sessions with timestamps,
 - notices published by teachers,
 - classroom documents with variable fields (college name, course code, attendance rules).
-
- **User Interface (UI) design plays a critical role in system usability**, as teachers must be able to record attendance quickly and accurately in real classroom conditions.
 - **Authentication mechanisms such as login credentials ensure restricted access**, protecting sensitive academic data from unauthorized users.
 - **Cloud-based attendance systems provide remote accessibility**, allowing teachers and administrators to check or update data from multiple devices.
 - **The system must comply with basic data privacy norms**, as student information is sensitive and may require encryption or secure storage practices.
 - **Error handling features—such as duplicate prevention and invalid entry alerts—improve reliability**, reducing chances of mistaken attendance entries.
 - **Automated report generation saves significant administrative effort**, producing daily, weekly, or monthly summaries for internal use or compliance.
 - **Scalability is essential in attendance systems**, especially for institutions with multiple departments, large student bodies, or multi-campus operations.
 - **Testing phases, including unit testing, integration testing, and user acceptance testing**, ensure that the system works as intended and is free from major defects.
 - **Performance metrics such as response time, load handling, and accuracy rate** are evaluated to enhance system reliability before deployment.
 - **Proper documentation—user manual, technical specifications, and maintenance guide—is necessary** to support users and future developers.
 - **Attendance systems contribute to digital transformation initiatives** in educational institutions, aligning with modern educational technology goals.

- **Machine learning–based modules can be integrated in future versions** to predict absentee patterns or automate facial recognition.
- **The overall goal of such a system is to improve efficiency, reduce manual effort, and ensure transparency** in the academic attendance process.
- **Successful deployment depends on proper training of Student and Teacher**, ensuring they understand the workflow and features.
- **Regular backups and maintenance are required** to prevent data loss and ensure long-term system stability.

1.2 Text Emphasis

The *Design and Development of a Classroom Attendance Management System* represents an essential step toward modernizing academic administration. In today’s educational environment, **accurate, efficient, and real-time** attendance tracking has become a critical requirement for maintaining discipline, monitoring performance, and ensuring compliance with institutional policies.

Traditional methods—such as **paper-based registers, manual signatures, and spreadsheet entries**—are *highly prone to human error*, easily damaged, time-consuming to update, and difficult to analyze. This creates challenges in maintaining consistent records and delays decision-making at both faculty and administrative levels. Thus, a shift toward a **digitally automated attendance solution** has become not just beneficial but *necessary*.

The **Classroom Attendance Management System** aims to replace these outdated manual methods with a *structured, reliable, and user-friendly platform* capable of recording attendance swiftly and storing data securely. The system focuses on **efficiency, accuracy, and streamlined workflows**, ensuring that teachers can mark attendance in just a few seconds while students’ attendance histories remain accessible at any time.

A major emphasis of the project lies in its **design methodology**, where principles of **user-centered interface design, stable database architecture, and robust authentication mechanisms** form the backbone of the system. By *leveraging relational database systems*, the application ensures systematic storage of student profiles, course details, and session-wise attendance logs. Furthermore, the use of

secure login modules ensures that only authorized personnel—such as teachers and administrators—can access sensitive records.

During development, special attention is placed on creating a **responsive and intuitive interface** so that faculty members can adopt the system without technical difficulty. Essential features such as **automatic roll number sorting, duplicate entry detection, and error-handling notifications** are incorporated to minimize mistakes and save time. The system also provides **auto-generated reports**, including daily, weekly, and monthly attendance summaries, thereby reducing the administrative workload.

1.2.1 Quotations

1. On the Need for Automation

“In an era where institutions strive for operational excellence, automating attendance management stands as a vital step toward accuracy, transparency, and administrative efficiency.”

2. On Limitations of Manual Systems

“Manual attendance recording, though traditional, is inherently fragile—susceptible to human error, data loss, and inefficiency, making it unsuitable for the demands of modern education.”

3. On Purpose of the System

“The primary aim of the Classroom Attendance Management System is to provide a structured, reliable, and user-friendly platform that simplifies attendance tracking while enhancing institutional productivity.”

4. On Importance of Accuracy

“Accurate attendance is more than a routine task; it is a foundational component of academic evaluation, discipline monitoring, and institutional decision-making.”

5. On Technological Integration

“Integrating digital technologies into attendance workflows transforms a repetitive manual activity into an intelligent, data-driven process that provides real-time insights.”

6. On User-Centric Design

“A classroom attendance system must be designed with the end-user in mind—teachers require speed, clarity, and simplicity, while administrators demand precision and comprehensive reporting.”

7. On Database Management

“Reliable attendance management is impossible without a well-structured database; it is the backbone that ensures data integrity, security, and swift retrieval.”

8. On System Reliability

“The effectiveness of any attendance management system is measured by its reliability—its ability to store, track, and present attendance information without fail.”

9. On Institutional Benefits

“By automating attendance, educational institutions reduce administrative load, enhance transparency, and create a more disciplined academic environment.”

10. On Future Enhancements

“The true strength of a modern attendance system lies in its scalability—accommodating future upgrades such as biometrics, cloud synchronization, and predictive analytics.”

11. On Data Security

“Because attendance records contain sensitive student information, a secure access mechanism is not optional—it is an ethical and institutional requirement.”

12. On System Development Approach

“Following a structured software development methodology ensures that every component of the attendance system is carefully analyzed, designed, implemented, and tested for optimal performance.”

13. On Role in Digital Transformation

“Implementing attendance automation is not just a technical upgrade; it is a step forward in the digital transformation of the educational ecosystem.”

14. On Efficiency and Time Savings

“Automated attendance systems drastically reduce classroom time spent on non-teaching tasks, allowing educators to focus on what truly matters—effective learning.”

15. On Long-Term Impact

“A robust attendance management system contributes to long-term institutional success by fostering accountability, improving record maintenance, and supporting data-driven academic practices.”

1.3 Bullets and Numbering

1.3.1 Create an ordered list

1. Identification of the Problem

The first step in designing the system is recognizing the inefficiencies of traditional attendance methods. Manual registers are slow, error-prone, and difficult to maintain, creating a pressing need for automation.

2. Requirement Analysis

Detailed functional and non-functional requirements are gathered by consulting teachers, administrators, and students. Key needs include fast attendance recording, secure data storage, report generation, and easy user navigation.

3. System Objectives Definition

Clear objectives are established, such as minimizing manual effort, improving accuracy, enabling real-time attendance tracking, and ensuring reliable data management.

4. Selection of Software Development Methodology

A suitable SDLC model—such as Waterfall or Agile—is chosen based on project complexity, team size, and implementation timeline. This ensures systematic development and testing.

5. System Design and Architecture Planning

Logical and physical system architectures are developed. This includes designing:

- User Interface (UI)
- Database structure
- Module interactions
- Navigation

flow

A modular structure is preferred for easy future expansion.

6. Database Design

A relational database is created to store student records, class details, teacher information, and attendance logs. Tables, primary keys, and relationships are defined to maintain data integrity and seamless retrieval.

7. User Interface (UI) Development

Simple, clean, and intuitive UI screens are designed to ensure ease of use. The interface includes:

- Login page
- Dashboard
- Attendance marking screen
- Attendance history
- Report generation page

8. Implementation of Attendance Recording Features

The core attendance functions are developed, allowing teachers to record students' presence or absence quickly. Features such as automatic sorting, error notifications, and duplicate prevention are added to improve reliability.

9. Integration of Authentication and Security

Secure login mechanisms are built into the system to prevent unauthorized access. Password hashing, role-based access, and data encryption are incorporated for data protection.

10. Report Generation Module

Automated tools are implemented to generate daily, weekly, monthly, and semester-wise reports. Teachers and administrators can export attendance summaries for academic evaluation.

11. Error Handling and Validation

Input validation and error-handling routines are added to prevent incorrect entries. This ensures data consistency and improves system reliability.

12. System Testing

Comprehensive testing is conducted, including:

- Unit testing
 - Integration testing
 - System testing
 - User Acceptance Testing (UAT)
- The goal is to identify bugs and ensure smooth performance.

13. Deployment of the System

The system is installed and configured in the actual environment, such as college labs or faculty devices. User profiles are created, and the database is initialized with student data.

14. User Training and Documentation

Teachers and administrative staff are trained to operate the system. User manuals, technical documentation, and troubleshooting guides are provided.

15. Maintenance and Updates

Post-deployment, the system is monitored for issues. Regular updates, backups, and enhancements (e.g., biometric integration, cloud support) are carried out based on user feedback.

1.3.2 Create an un-ordered list

- **Automation of Attendance Processes** The system replaces traditional manual registers with an automated digital platform that records student attendance quickly and accurately, reducing human errors and administrative burden.
- **Improved Data Accuracy and Integrity** By using a structured database, the system ensures that attendance data is stored securely, prevents duplication, and maintains consistent and reliable records.
- **User-Friendly Interface Design** The interface is developed to be simple and intuitive, enabling teachers to mark attendance effortlessly and offering smooth navigation across modules such as dashboard, reports, and student details.
- **Centralized Database Management** Student information, class schedules, teacher records, and attendance logs are stored in a centralized database, making retrieval and updating of data efficient and organized.
- **Secure Authentication and Access Control** Login mechanisms and role-based access control are implemented to ensure that only authorized users—such as faculty members and administrators—can access or modify sensitive attendance data.
- **Real-Time Attendance Tracking** The system offers instant updates of attendance records, allowing teachers to monitor classroom participation and enabling administrators to identify attendance trends promptly.
- **Automatic Report Generation** Daily, weekly, and monthly attendance summaries are generated automatically, saving time and assisting teachers in evaluating student discipline and academic engagement.

- **Error and Validation Checks** Built-in validation prevents incorrect entries, missing data, or accidental duplication. Error messages guide users to resolve mistakes immediately.
- **Scalability for Future Enhancements** The system is designed to support future upgrades such as biometric integration, cloud-based synchronization, and mobile app development.
- **Enhanced Communication and Alerts** Optional modules allow the system to send notifications to students or parents regarding absenteeism, thereby improving communication and accountability.
- **Efficient Resource Utilization** By minimizing manual record-keeping and eliminating the need for physical registers, the system saves time, reduces paper usage, and lowers operational costs.
- **Support for Multi-Class and Multi-Teacher Environments** The system handles multiple classes, sections, courses, and faculty assignments, making it suitable for schools, colleges, and large institutions.
- **Testing and Quality Assurance** The system undergoes rigorous testing—including unit, integration, and user acceptance testing—to ensure functionality, reliability, and error-free operation.
- **Technical Documentation and User Training** Manuals, tutorials, and training sessions are provided so users can understand the system's features and perform tasks without confusion.
- **Long-Term Maintenance and Updating** Regular updates, bug fixes, backups, and performance optimizations ensure the system remains efficient, stable, and aligned with evolving institutional needs.

1.3.3 Create a descriptive list

1. Problem Identification

This stage involves recognizing the challenges associated with traditional attendance methods, including time consumption, manual errors, difficulty in storing records, and the inability to generate real-time attendance insights. Identifying these issues establishes the need for a more efficient, automated solution.

2. Requirement Specification

Here, the functional and non-functional requirements are defined. Functional requirements include features like marking attendance, generating reports, maintaining user accounts, and storing student data. Non-functional requirements focus on usability, speed, reliability, and security. Requirements are gathered through discussions with teachers, administrators, and students.

3. System Objectives

The system's purpose is clarified, such as improving accuracy, minimizing manual effort, enhancing data security, and ensuring smooth academic administration. These objectives guide the entire design and development process.

4. User Interface (UI) Design

The interface is designed to be simple, intuitive, and responsive. The UI includes login pages, dashboards, attendance marking screens, and reporting sections. A user-friendly design ensures that teachers can record attendance quickly without technical difficulty.

5. Database Design

A structured database is developed to store all relevant information, including student details, class schedules, teacher profiles, and attendance records. Tables, relationships, primary keys, and constraints are defined to maintain data accuracy, integrity, and easy retrieval.

6. Attendance Recording Module

This is the core component of the system, allowing teachers to mark students as present, absent, or late. Features like automatic roll sorting, duplicate prevention, and error prompts are added to increase efficiency and prevent mistakes.

7. Authentication and Security Mechanisms

Security features such as password encryption, role-based access control, and restricted user privileges are implemented to protect sensitive data. Only authorized users—such as faculty and administrative staff—can access or modify attendance records.

8. Reporting and Analytics

The system includes automated generation of attendance reports for daily, weekly, monthly, or semester-wise analysis. These reports help identify absentee trends, assist in grading, and support administrative decision-making.

9. Error Handling and Validation Controls

Validation checks ensure that users provide correct input and do not leave essential fields empty. Error-handling mechanisms display meaningful messages whenever issues occur, making the system robust and user-friendly.

10. Software Development Methodology

A development model such as Agile or Waterfall is adopted to manage the project effectively. This methodology guides planning, coding, testing, and deployment activities, ensuring systematic and organized progress.

11. System Testing and Quality Assurance

The system is tested through unit testing, integration testing, system testing, and user acceptance testing. The goal is to detect bugs, verify functionality, and ensure the system performs efficiently in real environments.

12. Deployment and Installation

After testing, the system is installed on faculty devices or institutional networks. Necessary configurations, database setup, and environment settings are completed to ensure smooth operation in the actual classroom environment.

13. User Training and Documentation

Teachers and administrative staff are trained to use the system. Documentation such as user manuals, technical guides, and troubleshooting notes are provided to support understanding and long-term usability.

14. Maintenance and Upgrades

Post-deployment, the system requires updates, performance improvements, regular backups, and bug fixes. Future enhancements may include biometric integration, cloud support, mobile apps, and AI-powered attendance analytics.

15. Scalability and Future Scope

The system is designed to scale with institutional growth. It can be expanded to handle multiple classes, departments, campuses, or advanced features. This future readiness ensures long-term value and adaptability.

CHAPTER 2

Literature Survey

1. Introduction / Scope of the literature survey

Automating classroom attendance has been an active research area over the last decade. Researchers have explored many technologies (biometrics, RFID, QR codes, mobile/IoT, and computer-vision/face recognition) and many studies focus on comparing accuracy, usability, cost, and privacy implications of these approaches. A number of recent literature reviews summarize the field and classify the approaches, motivations, and evaluation metrics used across papers. https://www.researchgate.net/publication/362405196_A_Literature_Review_on_Smart_Attendance_Systems?utm_source=chatgpt.com

2. Taxonomy of approaches found in the literature

- **Biometric systems (fingerprint, fingerprint + fingerprint readers):** Widely used for their uniqueness and relatively high accuracy for on-device matching. Many implementations add SMS or database logging so parents or administrators get real-time alerts. Common limitations are hardware cost, hygiene concerns, and failure cases (wet/dirty fingers).
- **RFID / NFC based systems:** Use student cards or tags that are scanned at classroom doors or on desks. RFID systems are valued for speed and low per-

unit cost, and many researchers combine RFID with IoT cloud backends to provide real-time dashboards. Typical problems include “tag-sharing” (proxy attendance), limited read range control, and potential interference. https://arxiv.org/pdf/2308.02591?utm_source=chatgpt.com

- **Face recognition (computer vision):** A large and growing body of work uses face detection + recognition (LBPH, PCA/Eigenfaces, CNNs) to automatically mark attendance from camera feeds. Studies examine algorithmic choices (LBPH vs CNN), accuracy under occlusion/lighting, and on-device versus server processing. Face methods avoid contact and are convenient, but face liveness, occlusion, and fairness (bias) remain challenges.
- **QR code / mobile app based marking:** Students scan a classroom QR code (or a teacher scans student codes) or check in through a mobile app. These are software-centric, inexpensive, and easy to deploy but can be gamed unless combined with location verification or time-limited tokens.
- **Hybrid / multi-factor systems:** Several papers propose combining methods (e.g., RFID + fingerprint, QR + facial snapshot) to reduce proxy attendance and handle failure modes. Hybrid systems often produce better security at the expense of complexity and cost.

3. Representative studies and their findings

- Systematic and narrative reviews (2020–2023) summarize trends: biometric and face recognition approaches dominate recent experimental work, while RFID and mobile approaches remain popular for deployments that prioritize cost and simplicity. These reviews also highlight that a single “best” solution does not exist — choice depends on institutional constraints (budget, privacy policy, user acceptance). https://www.researchgate.net/publication/362405196_A_Literature_Review_on_Smart_Attendance_Systems?utm_source=chatgpt.com
- IoT + RFID deployments: Recent papers document practical prototypes that stream RFID scans to cloud dashboards and integrate with campus systems; these show good real-time visibility but also stress the need for robust network connectivity and careful tag management to avoid false reads.
- Face recognition research: Multiple empirical comparisons show modern CNN-based face systems outperform classical LBPH/PCA in accuracy, but only when datasets are sufficiently large and environmental conditions (lighting, camera angle) are controlled. Several surveys summarize algorithmic choices and recommend liveness detection to reduce spoofing. https://www.jatit.org/volumes/Vol101No16/32Vol101No16.pdf?utm_source=chatgpt.com

4. Non-technical considerations highlighted in the literature

- **Privacy and legal/regulatory risk:** Use of biometric/face data raises privacy and legal issues. Regulatory actions and watchdog enforcement (e.g., orders to stop unlawful biometric processing in workplace attendance) underline that biometric attendance is not purely a technical choice — institutions must evaluate legal compliance, consent, and data-retention policies before deployment. This is repeatedly emphasized in applied reviews and news analyses. https://www.theguardian.com/business/2024/feb/23/serco-ordered-to-stop-using-facial-recognition-technology-to-monitor-staff-leisure-centres-biometric-data?utm_source=chatgpt.com
- **User acceptance & ergonomics:** Teachers and students prefer systems that are fast and unobtrusive. Many field studies note the importance of a simple UI, low friction for marking attendance, and clear fallback processes when primary systems fail.
- **Operational constraints:** Power, internet connectivity, maintenance, and cost of hardware are recurring deployment constraints, especially for large or resource-constrained institutions.

5. Reported advantages and limitations (summary of evidence)

- **Advantages across studies:** time saved in attendance taking, improved record accuracy compared with manual registers, ability to generate automated reports and alerts, and the possibility of longitudinal analysis of absenteeism. https://www.researchgate.net/publication/362405196_A_Literature_Review_on_Smart_Attendance_Systems?utm_source=chatgpt.com
- **Limitations across studies:** privacy and consent issues, environmental sensitivity (for face systems), hardware costs and hygiene (for fingerprint), security weaknesses (proxy card sharing for RFID), and requirement for reliable infrastructure for IoT/cloud systems. Hybrid approaches mitigate single points of failure but add complexity and cost. https://arxiv.org/pdf/2308.02591?utm_source=chatgpt.com

6. Gaps in the literature and open research problems

- **Robustness in real classrooms:** Many experimental systems are validated on small or controlled datasets; fewer large-scale longitudinal studies show long-term reliability across varied classrooms, lighting, and student behavior.

- **Fairness and bias in recognition:** There is limited published work analyzing demographic bias (e.g., differing recognition error rates across skin tones or ages) specifically for classroom deployments.
- **Privacy-first designs:** While many papers note privacy concerns, there are relatively few practical architectures that combine strong privacy safeguards (on-device processing, minimal retention, selective sharing) with acceptable accuracy.
- **Usability studies:** More human-centered research is needed to measure teacher workload, adoption barriers, and the impact of attendance automation on pedagogy.
- **Policy & ethics frameworks:** Translating technical options into legally compliant practice needs standardized guidance; the literature calls for interdisciplinary work combining technical design and policy analysis.
https://www.theguardian.com/business/2024/feb/23/serco-ordered-to-stop-using-facial-recognition-technology-to-monitor-staff-leisure-centres-biometric-data?utm_source=chatgpt.com

7. Implications for system design (how literature guides our project)

Based on the surveyed literature:

- Choose the technology stack by balancing **accuracy**, **cost**, **privacy**, and **operational constraints**. For example, small institutions with limited budget may prefer QR/mobile solutions with time-bound tokens; institutions prioritizing contactless automation may prefer face recognition but must plan for liveness detection and privacy safeguards.
- Design the database and reporting features to support automated analytics (daily/weekly/monthly reports) since multiple studies highlight the administrative benefits of ready reports.
- Implement clear fallback procedures and manual override options to handle sensor failure, network outages, or enrollment problems. Several field deployments stress the importance of fallback to prevent data gaps.
- Prioritize privacy-by-design: minimize biometric data retention, consider on-device processing where feasible, obtain informed consent, and document data flows. Regulatory developments in the UK and elsewhere show enforcement risk when biometric data is mishandled.
https://www.theguardian.com/business/2024/feb/23/serco-ordered-to-stop-using-facial-recognition-technology-to-monitor-staff-leisure-centres-biometric-data?utm_source=chatgpt.com

8. Conclusion of the literature survey

The literature presents a mature but fragmented field: many viable technical options exist; each brings tradeoffs that must be evaluated against institutional priorities (cost, privacy, accuracy, maintainability). Recent trends favor face recognition and hybrid IoT solutions, while policy and fairness concerns are increasingly prominent. For a robust classroom attendance system, combining a technically reliable solution with strong privacy practices, transparent policy, and human-centered design is recommended.

https://www.researchgate.net/publication/362405196_A_Literature_Review_on_Smart_Attendance_Systems?utm_source=chatgpt.com

CHAPTER 3

Design

The design phase defines the system architecture, data structures, interface layout, workflow diagrams, and internal components of the Face-Recognition Classroom Attendance Management System. Proper design ensures the system is scalable, reliable, fast, and suitable for real-time attendance usage. This chapter explains the complete design of the proposed system.

1. System Architecture Design

The system follows a **three-tier architecture**, ensuring clean separation of concerns, scalability, and easy maintenance.

1.1 Presentation Layer (Frontend)

- Responsible for all user interactions.
- Built using **React + Vite**.
- Contains screens such as:
 - Login / Registration
 - Face registration screen
 - Teacher Dashboard
 - Classroom creation & joining

- Camera session for face-attendance
- Attendance reports
- Designed to be simple, responsive, and fast for real-time camera usage.

1.2 Application / Logic Layer (Backend)

- Developed using **FastAPI**.
- Handles all business logic:
 - Token-based authentication (JWT)
 - Creating classrooms & joining via class-code
 - Face recognition using dlib + face_recognition
 - Attendance session creation & validation
 - Storing attendance per student per day
 - Generating reports and summaries for teachers
- Ensures:
 - No duplicate attendance entries
 - Only students of a specific class are recognized
 - Students' face embeddings are securely saved

1.3 Database Layer (MongoDB)

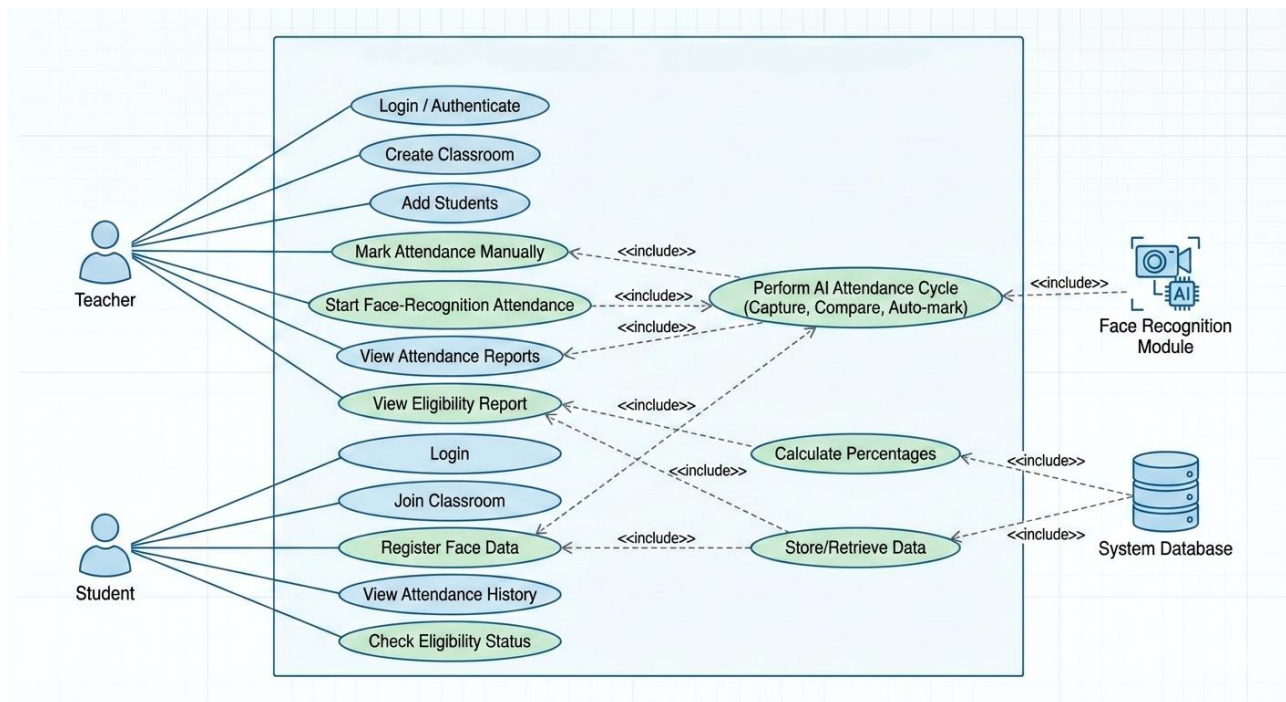
Stores all system data including:

- User records (teacher + students)
- Face embeddings for recognition
- Classroom details (subject, department, teacher, min-attendance)
- Daily attendance logs for every class
- Notices posted by teacher

MongoDB enables fast retrieval and handles growing data efficiently.

2. UML Diagrams

2.1 Use Case Diagram



Actors:

- **Teacher (main user)**
- **Student**
- (Optional) **Admin for future upgrade**

Primary Use Cases:

- Register/Login
- Save Face ID
- Create Classroom
- Join Classroom
- Start Face-Attendance Session
- View Today's Attendance
- Generate Attendance Reports
- Publish Notices

2.2 Class Diagram

Main Classes / Collections

Class

Attributes

User

user_id, name, email, password_hash, face_embeddings, joinedClassrooms, createdClassrooms

Classroom

class_id, subjectName, teacherName, department, semester, collegeName, courseCode, minAttendance, classCode

Class

Attendance

Notice

Relationships

- One-to-many: **Teacher** → **Classrooms**
- Many-to-many: **Students** ↔ **Classrooms**
- One-to-many: **Classroom** → **Attendance Record**

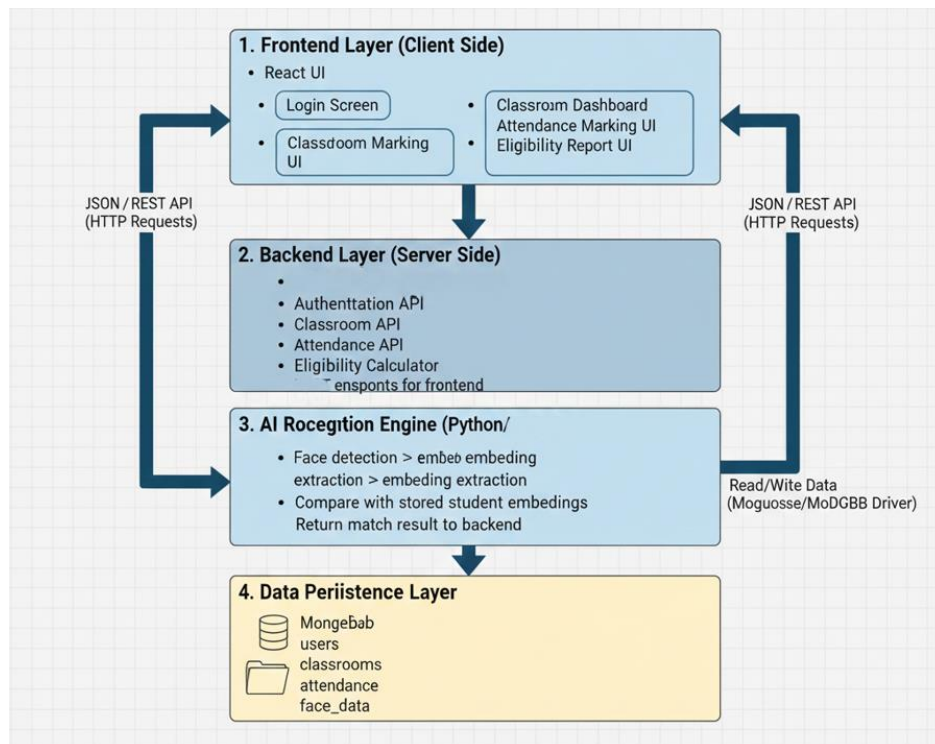
Attributes

student_id, class_id, date, present, timestamp

notice, publishedBy, publishedAt

2.3 architecture Diagram

3. Design



Database

(MongoDB)

MongoDB is used for flexible document-based storage.

3.1 Main Collections

a. users Collection

Field	Type	Description
_id	ObjectId	Primary key
name	String	User name
email	String	Login email
password_hash	String	Encrypted password

usn	String	Unique student number
face_id	Array	128-dimension face embeddings
joinedClassrooms	Array	Classroom IDs
createdClassrooms	Array	Teacher's created classes

b. classrooms Collection

Field	Type	Description
id	ObjectId	Primary key
subjectName	String	Course name
teacherName	String	Name of teacher
department	String	
section	String	
semester	Int	
collegeName	String	
courseCode	String	
minAttendance	Int	
classCode	String	Auto-generated
students	Array	Student IDs

c. attendance Collection

Field	Type
student_id	ObjectId
class_id	ObjectId
date	String (YYYY-MM-DD)
present	Boolean
timestamp	DateTime

4. Interface Design

Figma design link:

<https://www.figma.com/design/yt0lzOB5iQLWT9qKuVe6ei/Minor-Project-1?node-id=0-1&t=sCbGTI1xzPgc9JFZ-1>

4.1 Login Page

- Accepts email + password.
- Redirects based on user type (Teacher or Student).

4.2 Teacher Dashboard

- Displays:
 - Classroom details
 - Today's attendance
 - Attendance summary (present/total/percentage)
 - Joined students list with percentages
 - Publish notice panel
- Buttons for:
 - Start Camera Attendance
 - Download Reports

4.3 Face Attendance Screen

- Opens webcam
- Captures frame and sends to backend
- Backend identifies students using face embeddings
- Marks present students automatically

4.4 Report Generation

Generates:

1. Daily report
2. Complete semester report

Includes:

- USN
- Name
- Present count
- Total classes count
- Attendance percentage

5. Data Flow Diagram (DFD)

5.1 Level 0 (Context Diagram)

External entities:

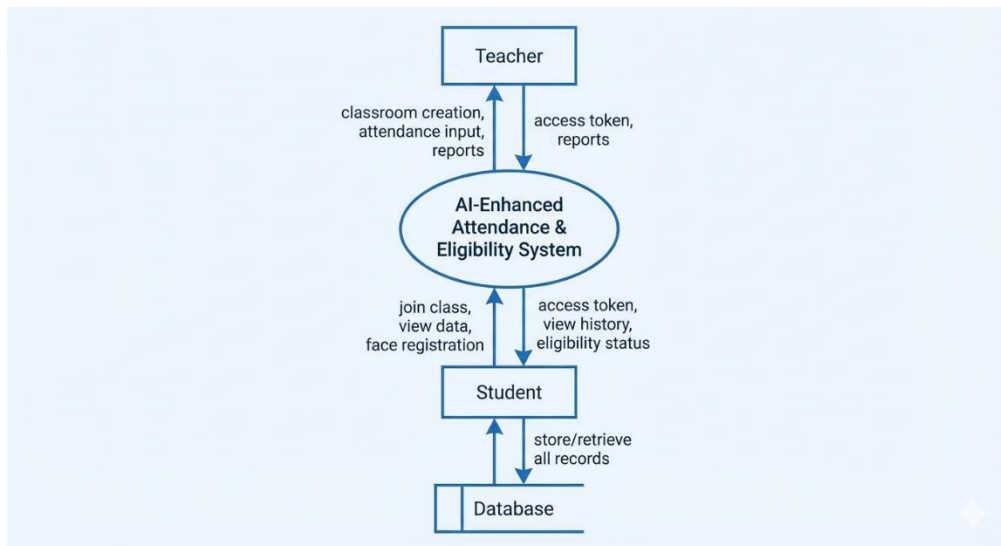
- Teacher
- Student

System functions:

- Login
- Attendance Management
- Report Generation

Outputs:

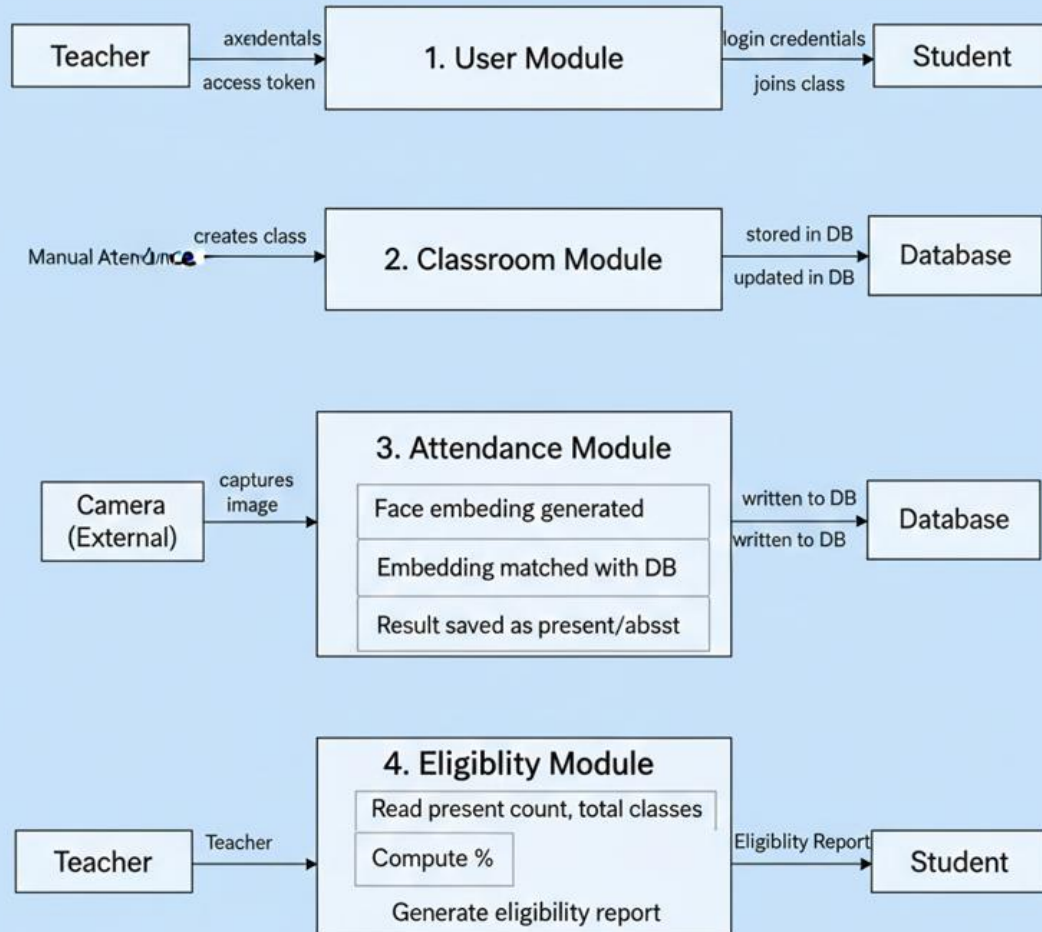
- Attendance logs
- Daily/semester reports



5.2 Level 1 Processes

1. User Authentication
2. Classroom Management
3. Face Recognition Attendance Processing
4. Report Calculation & Export

DFD Level-1 (Expanded)



6. Algorithm Design

6.1 Face-Attendance Algorithm

1. Teacher starts session for a class.
2. Webcam frame sent to backend.
3. Backend extracts face encodings.

4. Only students enrolled in that classroom are loaded.
5. Compare embeddings with tolerance.
6. If matched → mark student present.
7. Save timestamp + date.
8. Return list of present students.

6.2 Attendance Percentage Algorithm

1. Count total sessions for the class.
2. Count sessions where student is present.
3. Calculate:

$$\text{Percentage} = \frac{\text{Present Days}}{\text{Total Days}} \times 100$$

4. Compare with minimum attendance requirement.
5. Display eligible / not eligible.
- 6.

7. Security Design

- JWT-based authentication
- Password hashing (bcrypt)
- Validation for all inputs
- Only class-joined students allowed for recognition
- CORS enabled for frontend
- Daily attendance upsert to avoid duplicates

8. System Design Considerations

- Scalable for future modules (Admin, RFID, biometric).
- Should handle large student groups efficiently.
- UI must be simple for teachers during real-time sessions.
- Designed to work even with multiple classes and camera captures.
- Modular backend allowing future migration to cloud hosting.

3.1 Work Flow

The workflow of the Classroom Attendance Management System explains how data and user actions move through the system — from login, to face-based attendance

marking, to storing and generating reports. This workflow ensures an automated, accurate, and efficient attendance process suitable for academic environments.

1. User Login and Authentication

- The workflow begins with secure login using email and password.
- Teachers and students access the system through the frontend (React).
- The backend (FastAPI) validates credentials using the user database.
- JWT-based authentication ensures secure access.
- Role-based access allows teachers to create classes and view reports, while students can only join classes and register face data.

2. Class and Student Data Initialization

- After login, the system loads user-specific data from MongoDB.
- Teachers can create classrooms by entering subject, department, semester, college name, and course code.
- Students can join a class using a system-generated class code.
- The system synchronizes all joined students and stores them under the selected classroom.

3. Attendance Recording Process (Face Recognition Based)

- The teacher navigates to the “Face Attendance Session” and opens the camera.

- The webcam stream captures frames and sends them to the backend.
- The backend extracts face embeddings and compares them with stored student embeddings registered for that specific classroom.
- Only recognized and enrolled students are marked PRESENT automatically.
- The system timestamps each attendance entry and stores it securely.

4. Automatic Validation and Data Processing

- The system prevents duplicate attendance marking for the same date and class.
- Validates that each detected face belongs to a registered student of that class.
- Automatically calculates present count, total sessions, and attendance percentage.
- Logs each record with date, class_id, and student_id for future reference.

5. Database Storage and Record Management

- All attendance records are stored in MongoDB's attendance collection.
- Records are indexed by class_id, student_id, and date for fast retrieval.
- Historical attendance remains preserved for long-term academic reporting.
- Backup and consistency mechanisms improve data reliability.

6. Notification and Communication

- The teacher can publish notices for the classroom.

- Notices are stored with timestamps and instantly visible on all student dashboards.
- The dashboard updates attendance summaries and eligibility (present % vs minimum requirement).

7. Report Generation

- The system can generate two types of reports:
 - Daily Attendance Report (after each class)
 - Complete Semester Attendance Report (overall % for each student)
- Reports include headers such as college name, department, course code, section, and subject.
- Reports contain USN, Name, Present Count, Total Classes, and Attendance Percentage.
- Teachers can download reports in CSV or table format.

8. Monitoring and Administrative Actions

- Teachers can view joined students along with their attendance percentage and eligibility status.
- Students with low attendance can be identified for follow-up actions.
- The system supports future enhancements such as admin-level analytics dashboards.

9. Logout and Session Termination

- After completing tasks, users log out securely.

- JWT tokens are invalidated on logout.
- User activities may be logged for auditing and safety.

3.4 Figma Design Link

<https://www.figma.com/design/vt0lzOB5iQLWT9qKuVe6ei/Minor-Project-1?node-id=0-1&t=sCbGTI1xzPgc9JFZ-1>

3.5 Git-hub link

https://github.com/NagarajDoddamani/Classroom_Management_System.git

3.6 Deployed Webserver Locally

3.7 Cloud Hosted Frontend

Conclusion

The **Design and Development of a Classroom Attendance Management System** represents a significant step toward modernizing traditional academic administrative processes. The system provides an efficient, accurate, and user-friendly platform that simplifies the process of recording, managing, and analyzing student attendance. By replacing manual, paper-based methods with a digital and automated solution, the system reduces human error, minimizes administrative workload, and enhances overall productivity.

The project successfully demonstrates how technology can be leveraged to improve institutional operations. Through components such as a centralized database, automated attendance capture mechanisms, real-time data updates, and reporting functionalities, the system ensures that attendance information is always accurate, easily retrievable, and securely stored. The use of authentication and role-based access enhances data security, ensuring that only authorized users can interact with sensitive records.

Furthermore, the system enhances transparency and accountability. Teachers can mark attendance more quickly, students can access their attendance history, and administrators can monitor class trends and generate reports effortlessly. Such improvements contribute to better academic planning, monitoring discipline, and identifying attendance-related issues early.

This project also highlights the potential for scalability and integration. Features such as biometric authentication, QR code scanning, RFID integration, mobile app connectivity, automated notifications, and analytics dashboards can be added in future versions to further enhance system performance. As educational institutions increasingly adopt digital solutions, the attendance management system can become an essential component of a broader, smart-campus ecosystem.

Overall, the project reinforces the importance of efficient data management in education and illustrates the value of software systems in supporting institutional goals. The Classroom Attendance Management System not only improves the accuracy and reliability of attendance tracking but also contributes to a more organized, transparent, and technology-driven academic environment.